SMART CITY BIKE SHARING

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1. Project Overview

The dataset contains station metadata such as name, address, contract city and operational status information.Each record represents a station at a given last update timestamp with capacity and availability values.Geographic coordinates are provided to enable mapping and location based analytics.Boolean flags indicate optional station capabilities such as banking payment support and bonus status.The dataset can be used to monitor supply available bikes and demand free stands across cities.It supports operational reporting, service level monitoring and planning for station rebalancing.

2. Objective

Provide a structured understanding of bike sharing station capacity and availability by contract area.Enable descriptive reporting on station distribution capacity and operational status.Support diagnostic analysis to identify low availability or under utilized stations.
Prepare a clean dataset suitable for dashboards and predictive modeling use cases.
Generate actionable recommendations for rebalancing and maintenance prioritization.

3. Problem Statement

Bike sharing systems need continuous visibility into station level supply and docking availability.
Operational issues arise when stations frequently reach empty bike or full stand situations.
Raw station data often includes missing addresses mixed formats and inconsistent timestamps.
Stakeholders require a repeatable process to clean model and analyze station snapshots across cities.

4. Tools and Technologies

→ Microsoft Excel and Power Query for ingestion cleaning and transformations.
→ Power BI for interactive dashboards maps KPIs and contract level comparisons.

5. Data Preprocessing

- Set correct data types such as booleans for banking and bonus, whole numbers for capacities, and datetime for last update.
- Split the position field by comma into latitude and longitude and convert them into decimal numbers.
- Handle missing values by keeping blanks for address and validating numeric fields.

- Standardize status text to uppercase such as OPEN and CLOSED and trim whitespace in contract name.
- Create calculated columns such as occupancy rate bikes and availability rate stands.
- Remove duplicates by station number while keeping the latest last update per station.

6.Column Description

Column Name	Data Type	Description
station_number	INT	Unique identifier assigned to each bike station
contract_name	VARCHAR	Name of the city or contract under which the station operates
name	VARCHAR	Name of the bike station
address	VARCHAR	Physical location or address of the station
latitude	DECIMAL	Latitude coordinate of the station for geographic mapping
longitude	DECIMAL	Longitude coordinate of the station for geographic mapping
banking	BOOLEAN	Indicates whether the station supports card/online payments (TRUE/FALSE)
bonus	BOOLEAN	Indicates whether the station offers bonus incentives for bike returns
status	VARCHAR	Current operational status of the station (e.g., OPEN, CLOSED)
bike_stands	INT	Total number of bike stands available at the station
available_bikes	INT	Number of bikes currently available for use
available_bike_stands	INT	Number of empty stands available to park bikes
last_update	DATETIME	Timestamp of the most recent data update for the station
occupancy_rate	DECIMAL	Calculated ratio of available bikes to total bike stands
availability_rate	DECIMAL	Calculated ratio of available stands to total bike stands

7. Data Modeling

From: table (column)	Relationship	To: table (column)	Status
Fact_Bike_Status (Contract_ID)	*—◀—1	Dim_Contract (Contract_ID)	Active
Fact_Bike_Status (DateKey)	*—◀—1	Dim_Date (DateKey)	Active
Fact_Bike_Status (Station_ID)	*—◀—1	Dim_Station (Station ID)	Active
Fact_Bike_Status (status_id)	*—◀—1	Dim_Status (Status_ID)	Active

- The data model follows a **Star Schema architecture**, centering on a primary fact table, Fact_Bike_Status, which captures transactional bike data.
- Four specialized dimension tables—Dim_Contract, Dim_Date, Dim_Station, and Dim_Status—are utilized to provide descriptive context to the fact records.
- Each relationship is established as a **Many-to-One (*:1) cardinality**, ensuring that multiple fact records can map to a single unique attribute in a dimension.
- Primary keys such as Contract_ID, DateKey, Station_ID, and status_id serve as the relational anchors between the disparate data entities.

8.DAX:

```
Utilization % = DIVIDE ( [Docks Occupied], [Total Capacity (Stands)] )
```

```
Open Stations (Latest) =

VAR latest_ts = [Latest Timestamp]

RETURN

CALCULATE (

   DISTINCTCOUNT ( Fact_Bike_Status[station_id] ),

   Fact_Bike_Status[Last Update] = latest_ts,

   Dim_Status[status] = "OPEN"

)
```

```
Full Stations (Latest) =

VAR latest_ts = [Latest Timestamp]

RETURN

CALCULATE (

   DISTINCTCOUNT ( Fact_Bike_Status[station_id] ),

   Fact_Bike_Status[last update] = latest_ts,

   Fact_Bike_Status[available bike stands] = 0

)
```

```
Empty Stations (Latest) =

VAR latest_ts = [Latest Timestamp]

RETURN

CALCULATE (

   DISTINCTCOUNT ( Fact_Bike_Status[station_id] ),

   Fact_Bike_Status[Last Update]  = latest_ts,

   Fact_Bike_Status[Available Bikes] = 0

)
```

```
Bikes Available (Latest) =

VAR latest_ts = [Latest Timestamp]

RETURN

CALCULATE (

   [Bikes Available],

   Fact_Bike_Status[Last Update]= latest_ts

)
```

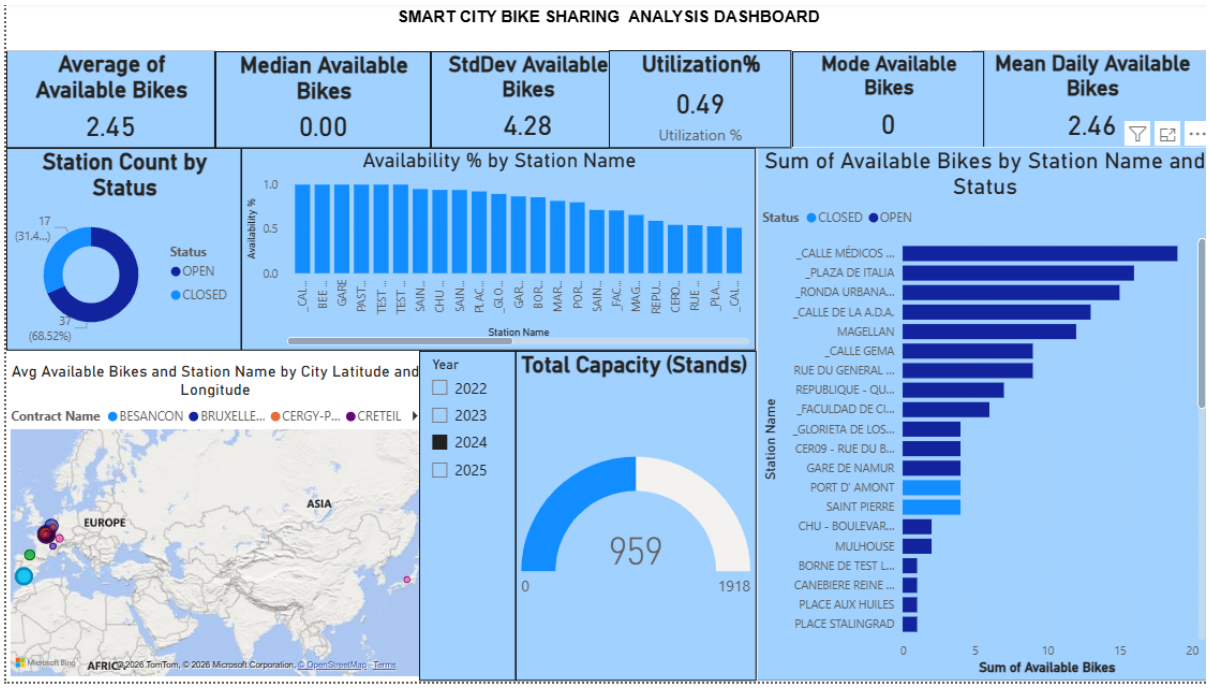
```
Station Utilization % (Latest) =

VAR latest_ts = [Latest Timestamp]

RETURN

CALCULATE (
```

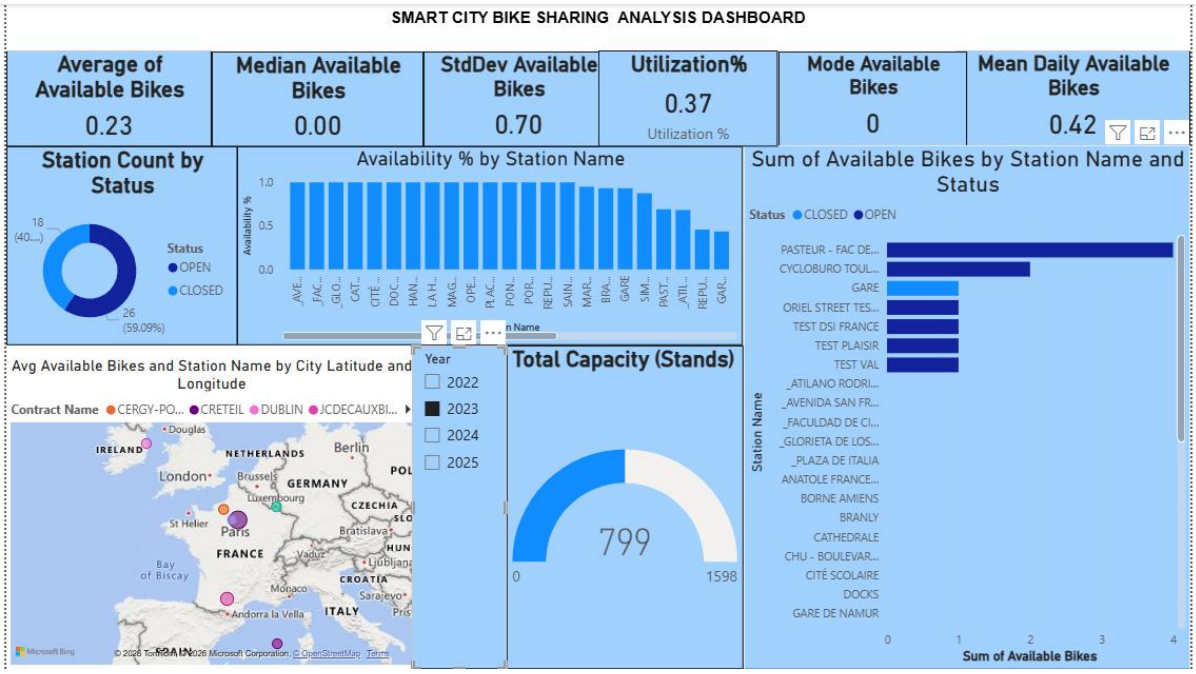
```
    DIVIDE (
        SUM ( Fact_Bike_Status[bike stands] ) - SUM ( Fact_Bike_Status[available bike stands] ),
        SUM ( Fact_Bike_Status[bike stands] )
    ),
    Fact_Bike_Status[last update] = latest_ts
)
```

```
% Stations Empty (Latest) =
DIVIDE ( [Empty Stations (Latest)], [Stations Count (Latest)] )
```

```
Availability % =
DIVIDE(
SUM(Fact_Bike_Status[Available Bike Stands]),
SUM(Fact_Bike_Status[Bike Stands]),
0
)
```

9.DASH BOARD


SMART CITY BIKE SHARING ANALYSIS DASHBOARD
Average of Available Bikes
2.45
Median Available Bikes
0.00
StdDev Available Bikes
4.28
Utilization%
0.49
Utilization %
Mode Available Bikes
0
Mean Daily Available Bikes
2.46
Station Count by Status
17 (31.4...)
37 (68.52%)
Status
OPEN
CLOSED
Availability % by Station Name
Availability %
Station Name
Sum of Available Bikes by Station Name and Status
Status CLOSED OPEN
Station Name
Sum of Available Bikes
Avg Available Bikes and Station Name by City Latitude and Longitude
Contract Name BESANCON BRUXELLE... CERGY-P... CRETEIL
EUROPE
ASIA
AFRICA
Year
2022
2023
2024
2025
Total Capacity (Stands)
959
0
1918


SMART CITY BIKE SHARING ANALYSIS DASHBOARD
Average of Available Bikes
0.23
Median Available Bikes
0.00
StdDev Available Bikes
0.70
Utilization%
0.37
Utilization %
Mode Available Bikes
0
Mean Daily Available Bikes
0.42
Station Count by Status
18 (40....)
26 (59.09%)
Status
OPEN
CLOSED
Availability % by Station Name
Availability %
Sum of Available Bikes by Station Name and Status
Status CLOSED OPEN
Station Name
Sum of Available Bikes
Avg Available Bikes and Station Name by City Latitude and Longitude
Contract Name CERGY-PO... CRETEIL DUBLIN JCDECAUXBI...
IRELAND
NETHERLANDS
GERMANY
FRANCE
ITALY
Year
2022
2023
2024
2025
Total Capacity (Stands)
799
0
1598

10. Analysis

Descriptive Analysis: What Happened

- Metric Summarization: Calculated total stations, total stands, and the count of available bikes and stands, segmented by contract name.
- Operational Status: Analyzed the station status breakdown to compare open versus closed units overall and by specific contracts.
- Resource Distribution: Evaluated the capacity distribution of bike stands and the real-time availability distribution of bikes across the network.
- Ranking Performance: Identified the top and bottom performing stations based on occupancy rates for bikes and availability rates for stands.
- Geospatial Visualization: Conducted geographic mapping of stations using latitude and longitude to visualize availability patterns across different regions.

Diagnostic Analysis: Why It Happened

- Availability Bottlenecks: Identified specific contracts with unusually low stand availability, pinpointing stations that are frequently full.
- Feature Correlation: Investigated differences in availability patterns between banking-enabled and non-banking stations.
- Behavioral Incentives: Assessed whether "bonus" stations exhibit different occupancy behaviors compared to standard stations.
- Outlier Investigation: Examined outlier stations characterized by high total capacity but consistently low numbers of available bikes.
- Data Integrity Validation: Compared open and closed stations while validating the accuracy of timestamps and geographic coordinates.

Predictive Analysis: What Will Happen

- Depletion Forecasting: Forecasted the probability of stations becoming completely empty based on current available bike trends.
- Availability Prediction: Predicted near-term availability by modeling contract-level station capacity and time-based features.
- Occupancy Modeling: Developed models to expect occupancy rates by contract to anticipate future rebalancing needs.

- Anomaly Detection: Implemented logic to detect anomaly risks associated with sudden, sharp drops in bike availability.
- Demand Hotspots: Estimated future demand hotspots using geographic clustering and historical availability trends.

Prescriptive Analysis: What Should Be Done

- Rebalancing Strategies: Recommended specific redistribution strategies to move bikes from high-availability to low-availability stations.
- Automated Alerting: Established alert thresholds for critical conditions, such as empty bike docks or completely full stand racks.
- Maintenance Prioritization: Prioritized maintenance schedules for stations frequently marked as closed or reporting outdated data.
- Capacity Optimization: Optimized long-term capacity planning based on observed usage trends and peak demand periods.
- Performance Monitoring: Developed comprehensive dashboards to monitor data freshness and key performance indicators (KPIs) in real-time.

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11. Conclusion

- The project provides a complete analysis of the bike-sharing system using descriptive, diagnostic, predictive, and prescriptive methods for better decision-making.
- It uncovers key insights into station performance, availability patterns, and resource distribution across different contracts.
- The analysis identifies operational inefficiencies, especially in stations with high demand and low availability.
- Predictive models enable forecasting of bike demand, station depletion risks, and future usage trends.
- Actionable recommendations such as bike rebalancing, maintenance prioritization, and alert systems improve overall efficiency.
- Overall, the project establishes a scalable, data-driven framework to enhance system performance and user satisfaction.